

KisanGat

UI / UX Design Brief

Smart Dairy Management & ERP System

Document Type	UI/UX Design Brief
Version	1.0 (MVP)
Platform	Web — React.js + Material UI (MUI)
Stack	React · Redux · Django REST · MySQL
Primary User	Single Admin — Dairy Owner
Date	June 2025

1. Design Philosophy

KisanGat serves a single, non-technical dairy owner who uses the platform every day to run their entire business. The design must feel **trustworthy, fast, and frictionless** — like a well-organized paper register, but digital. Every screen must communicate its purpose within 2 seconds of loading.

#	Principle	What It Means
1	Clarity over cleverness	Every number, action, and status instantly readable. No hidden menus, no ambiguous icons.
2	Data density with breathing room	Dairy operations involve many numbers. Show them clearly without cramming. Use whitespace.
3	Confidence through feedback	Every action (save, delete, generate) must respond with immediate visual confirmation.

■ The dairy owner is not a tech-savvy user. Every decision — from button placement to error messages — should reduce cognitive load and prevent mistakes.

2. Design Style

Primary Style: Soft UI Evolution (Enterprise SaaS variant)

Secondary Influence: Bento Grid for dashboard · Minimal & Swiss for data tables

Characteristic	Specification
Grid system	Strict 8px base grid throughout
Elevation	Soft shadows — never harsh borders for card separation
Corner radius	8–12px on cards, 8px on inputs, 100px on status badges
Motion	Functional only — 150–250ms transitions. No decorative animation.
Context	Agricultural roots acknowledged through earthy accent tones (not rustic)
Mode	Light mode primary. No dark mode in MVP.

Avoid:

- Glassmorphism — reduces readability on data-heavy screens
- Brutalism or flat-only — lacks the trustworthy, enterprise feel needed
- Heavy gradients or aurora backgrounds — distracting on dashboards
- Emoji as icons anywhere in the interface
- Decorative illustrations on operational screens (charts/tables are the visuals)

3. Color Palette

Primary Palette

	Token / Hex	Usage
	--color-primary #1B6CA8	Primary actions, active nav, links, key CTAs
	--color-primary-dark #145389	Hover/pressed state for primary elements
	--color-primary-light #E8F1F9	Selected row backgrounds, soft highlights
	--color-secondary #2E7D32	Success states, 'Paid' status, positive trends
	--color-secondary-light #E8F5E9	Success badge backgrounds
	--color-accent #F57F17	Warning states, 'Partial' payment, due alerts
	--color-accent-light #FFF8E1	Warning badge backgrounds
	--color-danger #C62828	Error states, 'Due/Unpaid' status, delete actions
	--color-danger-light #FFEBEE	Error badge backgrounds

Status Semantic Colors

Status	Background	Text Color	Border
Paid / Active	#E8F5E9	#2E7D32	#A5D6A7
Partial	#FFF8E1	#F57F17	#FFE082
Due / Inactive	#FFEBEE	#C62828	#EF9A9A

Neutral Palette

Token	Hex	Usage
--color-bg	#F5F7FA	App background (page canvas)
--color-surface	#FFFFFF	Cards, sidebar, modals
--color-surface-alt	#F0F4F8	Table row alternates, input backgrounds

Token	Hex	Usage
--color-border	#E2E8F0	Card borders, dividers, input outlines
--color-text-primary	#1A202C	Headings, labels, primary data
--color-text-secondary	#4A5568	Descriptions, secondary values
--color-text-muted	#718096	Timestamps, helper text, disabled labels

4. Typography

Use **Inter** as the single font family across the entire application. Inter has excellent tabular number support and is purpose-built for UI/dashboard contexts.

Token	Size	Weight	Line Height	Usage
--text-metric	32px	700	1.1	Dashboard KPI numbers (■12,450)
--text-display	28px	700	1.2	Page titles
--text-h1	22px	700	1.3	Section headings
--text-h2	18px	600	1.4	Card titles, modal titles
--text-h3	16px	600	1.4	Table column headers
--text-body	15px	400	1.6	Body text, descriptions
--text-label	14px	500	1.4	Form labels, nav items
--text-small	13px	400	1.5	Helper text, timestamps
--text-data	14px	400	1.4	Table cell values (tabular-nums)

■ Critical rule: Never use text smaller than 13px for any visible content. Numbers in tables must use font-variant-numeric: tabular-nums to prevent column width shifting. All monetary values must be prefixed with the ■ symbol.

5. Spacing & Layout System

Spacing Scale (8px base):

Value	Token	Usage
4px	--space-1	Icon + label gap, tight internal spacing
8px	--space-2	Component padding small
12px	--space-3	Input internal padding

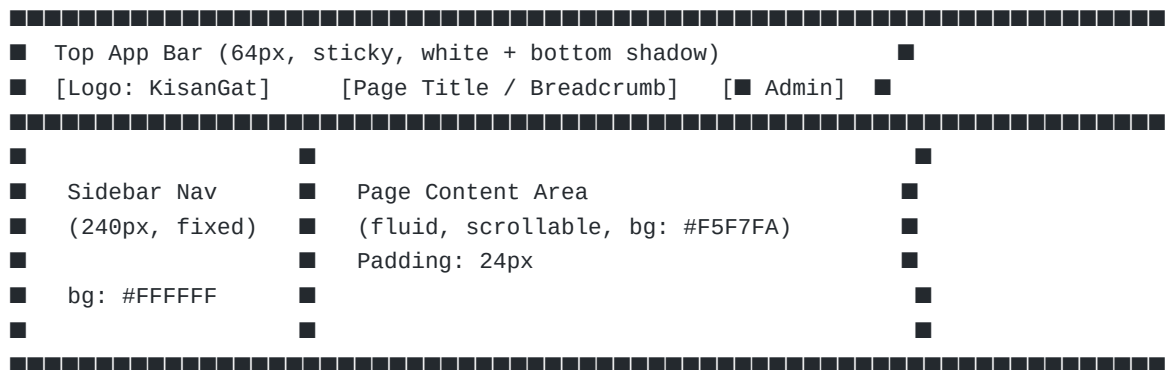
Value	Token	Usage
16px	--space-4	Card padding, list item padding
20px	--space-5	Section padding
24px	--space-6	Card gap, component separation
32px	--space-8	Section separation
48px	--space-12	Major layout sections

Responsive Breakpoints:

Breakpoint	Width	Layout Behavior
Mobile	375px	Single column · Sidebar = bottom sheet drawer · Tables → card stacks
Tablet	768px	Sidebar collapses to 64px icon rail · Content expands
Desktop	1024px	Full 240px sidebar · Standard dashboard grid
Wide	1440px+	Content max-width 1280px, centered · Larger chart areas

6. Layout Architecture

Global Shell



Sidebar Navigation

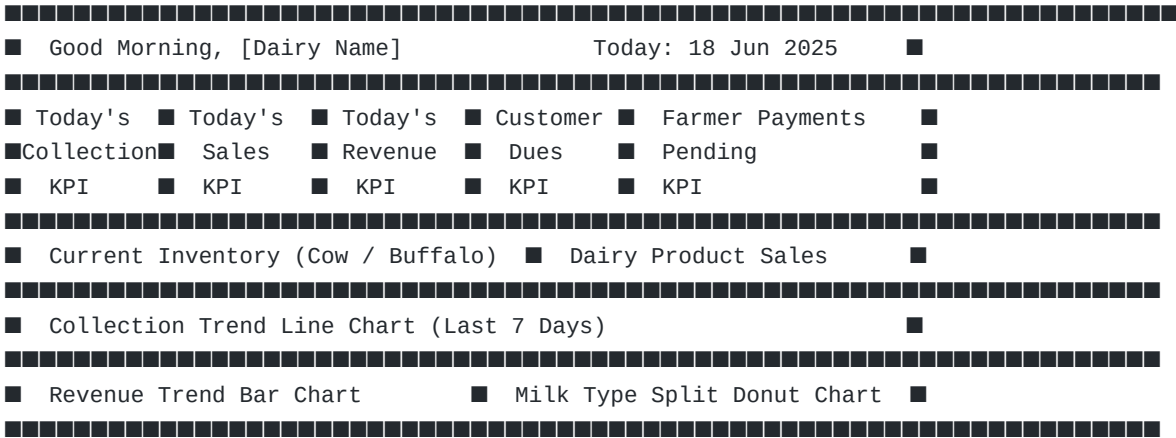
Element	Specification
Background	#FFFFFF with 1px right border (#E2E8F0)
Nav item height	48px with 16px horizontal padding
Icons	Lucide React, 20px, stroke-width 1.5
Active state	#E8F1F9 background + #1B6CA8 text + 3px left border accent

Element	Specification
Hover state	#F7FAFC background, 150ms transition
Group labels	11px, uppercase, --color-text-muted (e.g. OPERATIONS, FINANCE)
Logout	Visually separated by top border, danger color on hover

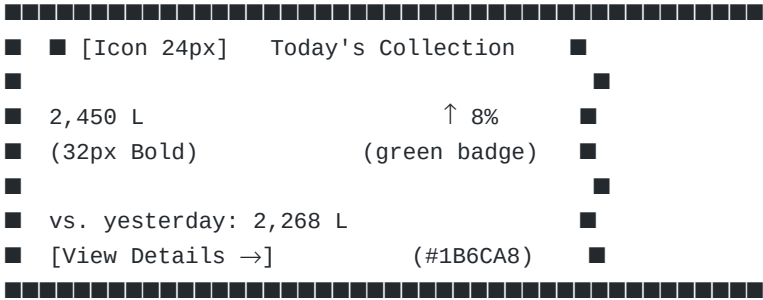
7. Dashboard Design

The dashboard is the most critical screen — it must communicate the entire business health at a glance. It is the first thing the dairy owner sees every morning.

Dashboard Layout (Desktop):



KPI Card Anatomy:



KPI Card Property	Specification
Background	White, border-radius 12px
Shadow	0 2px 8px rgba(0,0,0,0.06)
Icon container	40px circle, color-coded per metric
Metric number	32px, 700 weight, tabular-nums

KPI Card Property	Specification
Trend badge	Green ↑ or Red ↓, 12px, pill-shaped
Clickable	Entire card navigates to relevant module
Loading state	Shimmer skeleton — never blank card

Chart Specifications:

Chart	Type	Data	Color
Collection Trend	Line + area fill	Last 7 days, daily totals	--color-primary, 20% fill
Revenue Trend	Grouped bar chart	Cow / Buffalo / Products monthly	Primary / Secondary / Accent
Milk Type Distribution	Donut (2 segments)	Cow vs Buffalo split %	#1B6CA8 / #2E7D32

8. Component Design Specifications

8.1 Buttons

Type	Background	Text	Usage
Primary	#1B6CA8	White	Main actions: Save, Add, Generate
Secondary	White	#1B6CA8 border	Cancel, Back, secondary actions
Danger	#C62828	White	Delete — only in confirmation dialogs
Ghost	Transparent	Text-secondary	Tertiary actions

Property	Default	Compact (tables)	Mobile forms
Height	40px	36px	48px
Border-radius	8px	8px	8px
Font	14px / 500	13px / 500	14px / 500
Padding	0 20px	0 16px	0 24px

8.2 Form Inputs

Label Text (14px, weight 500, #1A202C) [Required *]

Input value or placeholder (muted, 14px) height: 44px

Helper text (13px, muted) OR Error message (13px, danger)

Property	Specification
Height	44px (meets 44pt touch target requirement)
Border default	1px solid #E2E8F0
Border focus	2px solid #1B6CA8 (no box-shadow outline)
Border error	2px solid #C62828
Border-radius	8px
Background	#FFFFFF (never gray on non-disabled)
Label position	Always above — never floating placeholder only
Date format	DD/MM/YYYY (Indian standard)
Default date	Today on Add Collection and Add Sale screens

8.3 Data Tables

[Page Title] [Search] [Filter] [+ Add New]									
Name	Type	Qty (L)	Amount	Status					
Row 1	Cow	12.5	450	Paid				white	
Row 2	Buffalo	8.0	480	Due				#FAFAFA	
Showing 1-20 of 134					[<] [1] [2] [3] [>]				

Table Element	Specification
Header row	#F7FAFC bg, 13px uppercase, text-muted, 500 weight
Row height	52px
Row hover	--color-primary-light bg, 150ms transition
Alternating rows	White / #FAFAFA
Numbers	Right-aligned, tabular-nums font feature
Dates	DD MMM YYYY format (e.g. 18 Jun 2025)
Action menu	Lucide MoreVertical, opens Edit / Delete dropdown
Pagination	MUI Pagination, 20 records per page default
Empty state	Centered illustration + message + primary CTA button
Sorting	Click header to toggle asc/desc with arrow icon

8.4 Status Badges

Status	Pill Style	Dot Color
Paid	#E8F5E9 bg · #2E7D32 text	● #2E7D32
Partial	#FFF8E1 bg · #F57F17 text	● #F57F17
Due	#FFEBEE bg · #C62828 text	● #C62828
Active	#E8F5E9 bg · #2E7D32 text	● #2E7D32
Inactive	#F0F4F8 bg · #718096 text	● #718096

Badge specs: padding 4px 10px · border-radius 100px · font 12px 500 weight · must include colored dot + text

8.5 Modals & Dialogs

Property	Specification
Backdrop	rgba(0,0,0,0.4)
Card	White, border-radius 16px, max-width 520px (form) / 400px (confirm)
Header	18px title, 600 weight + Lucide X close icon (top-right)
Footer	Right-aligned: [Cancel] [Primary Action]
Confirmation	Warning icon (amber) + red button for destructive actions
Close behavior	X button AND backdrop click closes the modal

9. Module-Specific UX Guidelines

9.x Milk Collection

- Default date = today. Never make the dairy owner manually select today's date.
- Amount auto-calculates from Quantity × Rate as the user types — no separate 'Calculate' button.
- Fat % field: step 0.1, range 3.0–8.0, show tooltip explaining fat-based rate adjustment.
- Farmer dropdown: searchable, shows Name + Mobile number in option.
- After saving: show success toast + clear form for next entry (don't redirect — dairy owner enters multiple collections at once).

9.x Milk Sales

- Payment status selector is required and must be the most prominent field after amount.
- When status = 'Partial', show additional field 'Paid Amount' (animate in smoothly).
- Inventory check: if sale quantity > remaining stock, show inline amber warning before save; block on submit.
- Customer ledger updates immediately after save.

9.x Customer Due Management

- Show summary banner at top: 'Total Outstanding: ■X across N customers.'
- Color-code rows by urgency: red >30 days overdue, amber 15–30 days, default <15 days.
- Record Payment modal: auto-fills max amount = outstanding balance, but allow partial.
- This is the most critical screen for daily cash flow — treat it with visual priority.

9.x Farmer Payment

- Date range selector prominent at top: Daily / Weekly / Monthly / Custom as tab pills.
- Auto-calculate total on date range selection — no manual trigger needed.
- 'Generate Payment' shows preview card first; 'Confirm & Mark Paid' finalizes it.
- Download Receipt as PDF — include dairy logo, period, itemized collection, total.

9.x Dashboard

- Load shimmer skeleton placeholders for all cards while data fetches (never blank cards).
- KPI cards are clickable and navigate to the relevant module.
- Empty state: 'Start by recording your first milk collection →' CTA button.
- Greeting changes by time: 'Good Morning / Good Afternoon / Good Evening, [Dairy Name]'.

9.x Billing

- Show bill preview in-app before download (iframe or rendered HTML).
- PDF includes: dairy logo, name/address, GST number, period, itemized records, total.

- Print-optimized CSS included for browser print (@media print).

10. Empty States

Every list/data screen must have a well-designed empty state. Never show an empty table frame with no content.

[SVG Illustration – 120px, muted colors, agriculture-themed]

No Farmers Added Yet
(16px, 600 weight, --color-text-primary)

Start building your farmer database to track milk
collection and payments accurately.
(14px, --color-text-secondary, max-width 320px, centered)

[+ Add First Farmer]
(Primary Button)

Use simple agriculture-themed SVG line illustrations with --color-primary-light fill and --color-primary stroke. Do NOT use generic 'box + magnifying glass' illustrations.

11. Mobile Responsiveness

Pattern	Mobile Adaptation
Data Tables	Each row becomes a card with field: value pairs. Actions as icon buttons.
Forms	Single column, full-width inputs. Sticky 'Save' button at bottom.
Dashboard KPIs	2-column grid (not 5-wide). Charts stack vertically.
Sidebar Navigation	Hidden behind hamburger → full-height left drawer with overlay.
Add actions	Floating Action Button (FAB) 56px at bottom-right, --color-primary.
Bottom padding	80px on all scrollable pages (mobile browser UI chrome clearance).
Touch targets	All interactive elements ≥ 44×44px.

12. Iconography

Icon Library: Lucide React (exclusively) · **Stroke Width:** 1.5px consistent

Module	Lucide Icon	Module	Lucide Icon
Dashboard	LayoutDashboard	Inventory	Package

Module	Lucide Icon	Module	Lucide Icon
Farmers	Users	Customer Dues	CreditCard
Milk Collection	Droplets	Farmer Payments	Wallet
Customers	UserCheck	Dairy Products	Box
Milk Sales	ShoppingCart	Employees	Briefcase
Product Sales	Tag	Expenses	Receipt
Bills	FileText	Reports	BarChart2
Settings	Settings	Logout	LogOut

Context	Size
Sidebar navigation	20px
Table action buttons	18px
Inside buttons	16px
Dashboard KPI cards	24px



13. Interaction & Motion

Interaction	Specification
Page transitions	Fade-in only, 150ms — no slide animations
Form save	Spinner on button (150ms delay before showing to prevent flicker)
Table row hover	Background color transition, 150ms ease
Modal open/close	Fade + scale 0.95→1.0, 200ms ease-out
Chart load	Animate drawing in, 400ms, only once on mount
Loading states	Always shimmer (not static gray) skeleton screens
Reduced motion	If prefers-reduced-motion enabled: remove all transitions and animations



14. Accessibility Requirements

Requirement	Specification
Color contrast	WCAG AA minimum: 4.5:1 body text, 3:1 large text
Focus rings	Visible 2px #1B6CA8 focus ring — never remove outline without replacement

Requirement	Specification
ARIA labels	All icon-only buttons must have aria-label attribute
Form labels	Every input must have associated <label> — no placeholder-only labels
Status badges	Must include text (Paid/Due/Partial) not color alone
Error messages	Must describe cause AND fix: not 'Error' but 'Quantity must be > 0'
Keyboard nav	All elements reachable via Tab, actions triggerable via Enter/Space
Toast ARIA	Use aria-live='polite' — toasts must not steal focus

15. Design Tokens — CSS Variables

```
:root {  
  /* Colors */  
  --color-primary:      #1B6CA8;  
  --color-primary-dark:  #145389;  
  --color-primary-light: #E8F1F9;  
  --color-secondary:     #2E7D32;  
  --color-secondary-light: #E8F5E9;  
  --color-accent:        #F57F17;  
  --color-accent-light:   #FFF8E1;  
  --color-danger:        #C62828;  
  --color-danger-light:   #FFEBEE;  
  --color-bg:            #F5F7FA;  
  --color-surface:        #FFFFFF;  
  --color-border:        #E2E8F0;  
  --color-text-primary:   #1A202C;  
  --color-text-secondary: #4A5568;  
  --color-text-muted:     #718096;  
  
  /* Typography */  
  --font-family: 'Inter', -apple-system, BlinkMacSystemFont, sans-serif;  
  --text-metric: 32px;    --text-display: 28px;  
  --text-h1:     22px;    --text-h2:     18px;  
  --text-h3:     16px;    --text-body:    15px;  
  --text-label:  14px;    --text-small:   13px;  
  
  /* Spacing */  
  --space-1: 4px;  --space-2: 8px;  --space-3: 12px;  --space-4: 16px;  
  --space-5: 20px; --space-6: 24px; --space-8: 32px;  --space-12: 48px;  
  
  /* Borders */  
  --radius-sm: 6px;  --radius-md: 8px;  
  --radius-lg: 12px; --radius-xl: 16px;  --radius-pill: 100px;  
  
  /* Shadows */  
  --shadow-sm: 0 1px 3px rgba(0,0,0,0.06);  
  --shadow-md: 0 2px 8px rgba(0,0,0,0.08);  
  --shadow-lg: 0 8px 24px rgba(0,0,0,0.12);  
}
```

```
--shadow-modal: 0 20px 60px rgba(0,0,0,0.15);

/* Layout */
--sidebar-width: 240px;
--sidebar-collapsed-width: 64px;
--topbar-height: 64px;

/* Transitions */
--transition-fast: 150ms ease;
--transition-base: 200ms ease;
--transition-slow: 300ms ease;
}
```

16. Visual Reference Analogies

These real-world products capture the intended look and feel for each design aspect:

Design Aspect	Reference Product	What to Borrow
Overall dashboard	Freshbooks	Clean KPI cards, structured data density
Sidebar navigation	Linear	Compact nav groups, active state treatment
Table design	Airtable	Row hover states, action menus, pagination
Status badges	Notion	Pill badge style, semantic color system
Form design	Stripe Dashboard	Label-above inputs, helper text, error clarity
Color tone	Zoho Books	Professional blue + green, practical not flashy
Mobile adaptation	Vyapar App	Card-stacked mobile tables familiar to Indian users

17. Pre-Delivery Checklist

Before any screen is considered complete, all items below must be verified. This checklist applies to every module.

Visual Quality

	Checklist Item
■	All numbers use tabular-nums font feature (no shifting column widths)
■	■ currency symbol shown before all monetary values
■	Dates shown in DD MMM YYYY format (e.g. 18 Jun 2025)
■	No emoji used anywhere as icons or decorations

■	Lucide React icons only, 1.5px stroke, consistent sizes
■	Status badges are pill-shaped with colored dot + text
■	Every list screen has an empty state: illustration + message + CTA

Interaction & Feedback

	Checklist Item
■	All buttons have hover, active, loading, and disabled states
■	cursor: pointer on all clickable elements
■	Form submission shows loading spinner on button
■	Success/error toasts appear after every action
■	Delete actions require confirmation dialog with danger button
■	Modal backdrop click closes the modal

Data & Forms

	Checklist Item
■	Amount fields auto-calculate where specified (no manual Calculate button)
■	Default date = today on collection/sales screens
■	Required field validation shown on submit (not on every keystroke)
■	Error messages state cause + how to fix (not just 'Required')
■	All dropdowns are searchable when options > 5

Responsiveness

	Checklist Item
■	Tested at 375px, 768px, 1024px, 1440px breakpoints
■	Tables convert to card-stacks on mobile
■	No horizontal scroll at any breakpoint
■	FAB present on mobile list screens
■	Touch targets $\geq 44 \times 44$ px on all interactive elements

Accessibility

	Checklist Item
■	Color contrast ≥ 4.5:1 for all body text (verified with tool)
■	Focus rings visible on keyboard navigation
■	All icon-only buttons have aria-label attribute
■	Status communicated by text, not color alone
■	Form errors use aria-live regions for screen reader announcement

This document is the single source of truth for all visual and UX decisions in KisanGat Version 1.0 MVP. All deviations from this brief must be explicitly documented and approved. When in doubt, prioritize clarity, consistency, and the needs of the daily dairy operator over aesthetic trends.